

Question: 5 (Marks: 1)

The magnitude of $\exp(2+3i)$ is _____.

Choices:

☐ $\exp(2)$

☐ $\exp(3)$

☐ $\exp(2*3)$

☐ $\exp(2/3)$

Complex Analysis and Differential Geometry

Question: 3 (Marks: 1)

$$f(z) = \frac{1}{z^2 + 1}$$

The function _____ is defined everywhere in the finite plane except at the points

Choices:

☐ $z = \pm 1$

☐ $z = \pm 2$

☐ $z = \pm i$

☐ $z = \pm 2i$

Complex Analysis

Question: 3 (Marks: 1)

If $z = e^{i\theta}$, then $z^n =$ _____

Choices:

☐ $z = \pm 1$

☐ $z = \pm 2$

☐ $z = \pm i$

☐ $z = \pm 2i$

Question: 4 (Marks: 1)

If $z = e^{i\theta}$, then $z^n =$ _____

Choices:

☐ $e^{i\theta}$

☐ $e^{-i\theta}$

☐ $e^{in\theta}$

☐ $e^{-in\theta}$

Complex Analysis and Differential Geometry

Question: 2 (Marks: 1)

The absolute value of the number, $z = 1 - i$ is _____

Choices:

2



$\sqrt{2}$



0



1



Question: 9 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $V_z =$ _____

Choices:

☐ -1☐ 1☐ 0☐ 2

Question: 8 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $U_z =$ _____

Choices:

☐ -1☐ 1☐ 0☐ 2

Question: 7 (Marks: 1)

Let $f(z) = \frac{1}{z}$ is discontinuous at $z = \underline{\hspace{2cm}}$

Choices:

0



1



-1

 i 

TIME LEFT

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Question: 6 (Marks: 1)

Square of $-2-5i$ is _____

Choices:

$-21 + 20i$



$-2-5i$



$-20-10i$



$20+10i$



Question: 9 (Marks: 1)

In the complex valued function $z = x - iy$, the value of $J'_z =$ _____

Choices:

☐ -1☐ 1☐ 0☐ 2

Complex Analysis and Differential Geometry

Question: 10 (Marks: 1)

The Polar Cauchy-Riemann equations on a pair of real-valued functions of two real variables u and v are _____.

Choices:

☐ $rU_r = V_\theta$ and $U_\theta = -rV_r$

☐ $U_r = V_\theta$ and $U_\theta = -V_r$

☐ $rU_\theta = V_r$ and $U_r = -rV_\theta$

☐ $U_r = V_\theta$ and $U_\theta = V_r$

Question: 11 (Marks: 1)

In the complex valued function $f(z) = \frac{1}{re^{i\theta}}$, the value of $U_r =$ _____

Choices:

☐ $\frac{1}{r}$

☐ $\frac{\cos \theta}{r}$

☐ $\cos \theta$

☐ e^{θ}

Question: 11 (Marks: 1)

$$f(z) = \frac{1}{re^{i\theta}}$$

, the value of $U_1 =$ _____

In the complex valued function

Choices:

☐ $\frac{1}{r}$

☐ $\frac{\cos \theta}{r}$

☐ $\cos \theta$

☐ $e^{i\theta}$

Question: 15 (Marks: 1)

Let $f(z) = 2z^3 + 2z - 7$, then $f'(z)$ is _____.

Choices:

☐ $6z^2 + 2$

correct

☐ $z^3 + 2$

☐ $2z^2 - 7$

☐ $2z^3 + 2$

TIME LEFT

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Complex Analysis and Calculus

Question: 14 (Marks: 1)

Let $f(z) = 3z^2 + 5z + 10$, then $f'(z)$ is _____

Choices:

☐ $5z$

☐ $6z$

☐ $6z - 5$

☒ $6z + 5$

correct

TIME LEFT

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Question: 13 (Marks: 1)

The limit of $f(z) = z^2 + 3z - 1$ at $z = i$ is _____

Choices:

☐ $3i$

☐ $3i-2$

☐ $2i-3$

☐ $2i-3$

Question: 12 (Marks: 1)

The limit of $f(z) = z^2 + z + 1$ at $z = i - 1$ is _____

Choices:

☐ 1☐ -1☐ i☐ -i

Complex Analysis and Differential Geometry

Question: 28 (Marks: 3)

Use Cauchy-Riemann equations to show that $f'(z)$ does not exist at any point if $f(z) = z - \bar{z}$, where $z = x + iy$.

Answer:

△△079 555555 555555 555555

TIME LEFT

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Question: 24 (Marks: 1)

If $z = a + ib$ is a complex number, then _____

Choices:

☐ $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$

☐ $\operatorname{Re}(z) = \frac{z - \bar{z}}{2}$

☐ $\operatorname{Re}(z) = \frac{z \bar{z}}{2}$

☐ $\operatorname{Re}(z) = z + \bar{z}$

correct

Question: 20 (Marks: 1)

Let $z = 1 + i$, then exponential form of z is _____

Choices:

☐ $z = \sqrt{3} e^{\frac{\pi i}{4}}$

☐ $z = e^{\frac{\pi i}{4}}$

☐ $z = \sqrt{2} e^{\frac{\pi i}{4}}$

☐ $z = \sqrt{2} e^{\frac{\pi i}{4}}$

correct

Complex Analysis and ...

Question: 16 (Marks: 1)

Let $f(z) = 2z^2 - 7z + 1$, then $f'(-2i)$ is _____

Choices:

☐ $-2i+1$

☐ $-4i-7$

☒ $-8i-7$

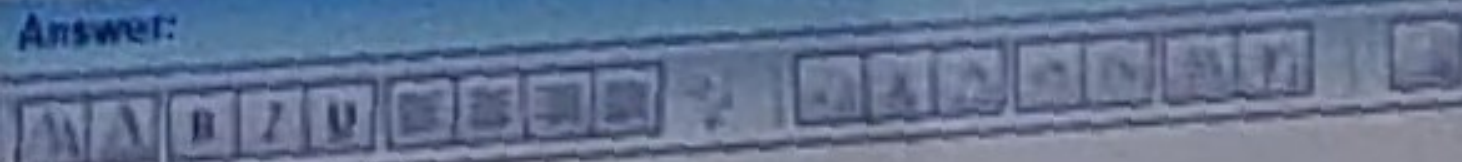
☐ $-8i+1$

correct

Question: 29 (Marks: 3)

Express the complex number $-\sqrt{3} + i$ in exponential form.

Answer:



Question: 25 (Marks: 1)

Which of the following are important part(s) of a function?

Choices:

☐ domain and range

☒ domain and rule

correct

☐ range and rule

☐ rule only

Question: 21 (Marks: 1)

The function $f(z) = \frac{1}{3z+i}$ is discontinuous at $z =$ _____

Choices:

☐ $3i$

☐ $-3i$

☐ $\frac{1}{3}$

☒ $-\frac{1}{3i}$

correct

A closed disc with center z_0 and radius ε is defined by _____

Choices:

☐ $\{z \in \mathbb{C} : |z - z_0| \geq \varepsilon\}$

☐ $\{z \in \mathbb{C} : |z + z_0| \geq \varepsilon\}$

☒ $\{z \in \mathbb{C} : |z - z_0| \leq \varepsilon\}$

correct

☐ $\{z \in \mathbb{C} : |z + z_0| \leq \varepsilon\}$

Question: 26 (Marks: 1)

If z lies inside the unit circle, then the image of z under the reciprocal function lies _____

Choices:

☐ on the unit circle

☐ outside the unit circle

☐ inside the unit circle

☐ nowhere in complex plane

Complex Analysis
Question: 22 (Marks: 1)

The function $f(z) = \frac{1}{z-3i}$ is discontinuous at $z =$ _____.

Choices:

☒ $3i$

correct

☐ $-3i$

☐ i

☐ $-i$

_____ was first to introduce notation of $i = \sqrt{-1}$.

Choices:

☐ Cardano

☒ Euler

☐ Rafael Bombelli

☐ Pythagoras

L.Euler
introduce this
notation and
Cardano
introduce
complex
number

correct

Question: 17 (Marks: 1)

A closed disc with center z_0 and radius ε is defined by _____

Choices:

☐ $\{z \in \mathbb{C} : |z - z_0| \geq \varepsilon\}$

☐ $\{z \in \mathbb{C} : |z + z_0| \geq \varepsilon\}$

☒ $\{z \in \mathbb{C} : |z - z_0| \leq \varepsilon\}$

correct

☐ $\{z \in \mathbb{C} : |z + z_0| \leq \varepsilon\}$

Complex Analysis and Differential Geometry

Question: 26 (Marks: 1)

If z lies inside the unit circle, then the image of z under the reciprocal function lies _____

Choices:

- ☐ on the unit circle
- ☐ outside the unit circle
- ☐ inside the unit circle
- ☐ nowhere in complex plane

Question: 27 (Marks: 1)

$$\lim_{z \rightarrow i} \frac{i z^3 - 1}{z + i} = \underline{\hspace{2cm}}$$

Choices:

1

 i 

0



-1

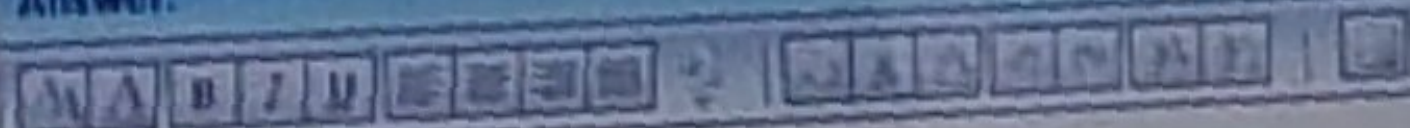


Question: 31 (Marks: 5)

If a complex number z satisfies $|z| < 2$, then find upper bound of

$$|z^3 + 3z^2 - 2z + 1|$$

Answer:



Suppose that $f(z_0) = g(z_0) = 0$, $f'(z_0)$ and $g'(z_0)$ exist, where $g'(z_0) \neq 0$. By definition $\frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x}$, show that $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$.

Answer:

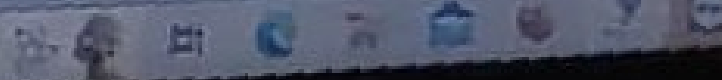
$\frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$

TIME LEFT

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Question: 23 (Marks: 1)

$$\frac{d}{dz} (20 z^5) = 20 \cdot 5 (z^{5-1}) \\ = 100 (z^4)$$

Choices:

☐ $10 z^4$

☐ $20 z^5$

☐ $50 z^4$

☐ $100 z^4$

correct

Question: 22 (Marks: 1)

The function $f(z) = \frac{1}{z-3i}$ is discontinuous at $z =$ _____.

if $z=3i$ then
 $f(z)=1/3i-3i=1/0$
 $= \text{infinity}$

Choices:

☐ $3i$

correct

☐ $-3i$ ☐ i ☐ $-i$

Question: 21 (Marks: 1)

The function $f(z) = \frac{1}{3z+i}$ is discontinuous at $z =$ _____.

$z = -i/3$ then

$$f(z) = 1/3 * -i/3 + i$$

$$= 1/-i + i = 1/0 =$$

infinity

Choices:

☐ $3i$

☐ $-3i$

☐ $\frac{1}{3} + i$

☐ $\frac{1}{3} - i$

correct

Question: 25 (Marks: 1)

Which of the following are important part(s) of a function?

Choices:

☒ domain and range

correct

☐ domain and rule

☐ range and rule

☐ rule only

Question 1 (Marks: 1)

If $z_1 = 2 + i$, $z_2 = 1 + 3i$, then $\operatorname{Re}(z_1 - z_2) = \underline{\hspace{2cm}}$

Choice

☐ 1☐ i ☐ $2i$ ☐ 2

Question 2 (Marks: 1)

The argument of the number, $-1+i$ is _____.

Choice

☐ 45° ☐ 90° ☐ 135° ☐ 180°

$$f(z) = \frac{1}{z^2 + 4}$$

The function _____ is defined everywhere in the finite plane except at the points _____.

Choice

☐ $z = \pm i$

☐ $z = \pm 2$

☐ $z = \pm 2i$

☐ $z = \pm 1$

Complex Analysis and Differential Geometry
Question 5 (Marks: 1)

$$\lim_{z \rightarrow \infty} \frac{z^2 + 1}{z - 1} =$$

Choice

☐ π

☐ i

☐ $-i$

☐ ∞

Question 4 (Marks: 1)

A function of a complex variable is said to be continuous in a region C if it is continuous at _____

Choice

☐ each point in C

☐ one point in C

☐ two points in C

☐ nowhere in C

Question 4 (Marks: 1)

A function of a complex variable is said to be continuous in a region C if it is continuous at _____.

Choice

☐ each point in C

☐ one point in C

☐ two points in C

☐ nowhere in C

Question 8 (Marks: 1)

Let $f(z) = \frac{1}{z}$ is discontinuous at $z = \underline{\hspace{2cm}}$.

Choice

☐ 0

☐

☐ 1

☐

☐ -1

☐

☐ i

☐

Question 7 (Marks: 1)

Square of $4-i$ is _____.

Choice

☐ $4-i$

☐ $15-8i$

☐ $8-2i$

☐ $8+2i$

The magnitude of $\exp(-2+3i)$ is _____.

Choice

☐ $\exp(-2*3)$

☐ $\exp(2*3)$

☐ $\exp(3)$

☐ $\exp(-2)$

Question 9 (Marks: 1)

The Cauchy-Riemann equations on a pair of real-valued functions of two real variables $u(x, y)$ and $v(x, y)$ are _____

Choice

☐ $U_x = V_y$ and $U_y = -V_x$

☐ $U_x = V_x$ and $U_y = -V_y$

☐ $U_y = V_y$ and $U_x = -V_x$

☐ $U_x = -V_y$ and $U_y = -V_x$

In the complex valued function $z = x - iy$, the value of $V_z =$ _____

Choice

-1

☐

1

☐

0

☐

2

☐

The Polar Cauchy-Riemann equations on a pair of real-valued functions of two real variables u and v are _____.

Choice

☐ $rU_r = V_\theta$ and $U_\theta = -rV_r$

☐ $U_r = V_\theta$ and $U_\theta = -V_r$

☐ $rU_\theta = V_r$ and $U_r = -rV_\theta$

☐ $U_r = V_\theta$ and $U_\theta = V_r$

The limit of $f(z) = \frac{1}{z^2 + 4}$ at $z = 1$ is _____.

Choice

4



1/3



1/4



1/5



Question 13 (Marks: 1)

The limit of $f(z) = 5z - 10$ at $z = i + 1$ is _____.

Choice

☐ $3i - 3$

☐ $5i - 5$

☐ $5i - 3$

☐ $3i - 5$

$$f(z) = \frac{1}{re^{i\theta}}$$

in the complex valued function

, the value of $U_r =$ _____

Choice

☐ $\frac{1}{r}$

☐ $\frac{\cos \theta}{r}$

☐ $\cos \theta$

☐ $e^{i\theta}$

Question 16 (Marks: 1)

Let $f(z) = z^3 - 3z + 12$, then $f'(z)$ is _____.

Choice

☐ $z^2 - 3$

☐ $3z^2 - 3$

☐ $z^2 - 1$

☐ $z^3 - 3$

Question 15 (Marks: 1)

Let $f(z) = 3z^2 + 5z + 10$, then $f'(z)$ is _____.

Choice

☐ $5z$

☐ $6z$

☐ $6z - 5$

☐ $6z + 5$

Question 15 (Marks: 1)

Let $f(z) = 3z^2 + 5z + 10$, then $f'(z)$ is _____.

Choice

☐ $5z$

☐ $6z$

☐ $6z - 5$

☐ $6z + 5$

Question 18 (Marks: 1)

Let $f(z) = 2z^2 - 7z + 1$, then $f'(-2i)$ is _____.

Choice

☐ $-2i+1$

☐ $-4i-7$

☐ $-8i-7$

☐ $-8i+1$

Let $f(z) = 2z^3 + 2z - 7$, then $f'(z)$ is _____.

Choice

☐ $6z^2 + 2$

☐ $z^2 + 2$

☐ $2z^2 - 7$

☐ $2z^3 + 2$

Which of the following statement is TRUE?

Choice

☐ The set R of real numbers is an extension of set C of complex numbers.

☐ The set C of complex numbers is an extension of set R of real numbers.

☐ The set Q of rational numbers is an extension of set Q' of irrational numbers.

☐ The set Q' of irrational numbers is an extension of set Z of integers

The complex number $z = x + iy$ is real if and only if _____.

Choice

☐ $x = 0$

☐ $y = 0$

☐ Both x and y must be 0.

☐ y must not be 0.

If $|z| = |3 + i4| = 5$, then $|\bar{z}| =$ _____.

Choice

3

☐

4

☐

5

☐

6

☐

Question 21 (Marks: 1)

If $|z| = |3 + i4| = 5$, then $|\bar{z}| =$ _____.

Choice

3

☐

4

☐

5

☐

6

☐

Questio 24 (Marks: 1)

The function $f(z) = \frac{1}{z^2 + 25}$ is discontinuous at $z = \underline{\hspace{2cm}}$.

Choice

☐ $\pm 2i$ ☐ $\pm 3i$ ☐ $\pm 4i$ ☐ $\pm 5i$

$$\frac{1}{z_1} = \frac{1}{r_1 e^{i\theta_1}}$$

Let $z_1 = r_1 e^{i\theta_1}$ be a point inside the unit circle, then the reciprocal $\frac{1}{z_1}$ will lie

Choice

☐ inside the unit circle.

☐ outside the unit circle.

☐ on the unit circle.

☐ nowhere in complex plane.

Question 22 (Marks: 1)

The function $f(z) = \frac{1}{2z-i}$ is discontinuous at $z = \underline{\hspace{2cm}}$.

Choice

☐ i ☐ $-i$ ☐ $\frac{i}{2}$ ☐ $-\frac{i}{2}$

The punctured open disc with radius ε and center at z_0 is defined by _____

Choice

☐ $\{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$

☐ $\{z \in \mathbb{C} : |z - z_0| \geq \varepsilon\}$

☐ $\{z \in \mathbb{C} : 0 < |z - z_0| < \varepsilon\}$

☐ $\{z \in \mathbb{C} : |z - z_0| > \varepsilon\}$

The expression $(\cos \theta + i \sin \theta)^3$ can be written as _____.

Choice

☐ $3\cos \theta + i 3 \sin \theta$

☐ $\cos 3\theta + i \sin 3\theta$

☐ $3\cos \theta + i \sin \theta$

☐ $3\cos 3\theta + i 3 \sin 3\theta$

$f(z) = \frac{1}{z} = z^{-1}$ is a polynomial.

Choice

☒ True

☐ False

The punctured open disc with radius ε and center at z_0 is defined by _____

Choice

☐ $\{z \in \mathbb{C} : |z - z_0| < \varepsilon\}$

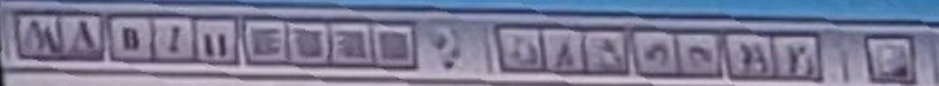
☐ $\{z \in \mathbb{C} : |z - z_0| \geq \varepsilon\}$

☐ $\{z \in \mathbb{C} : 0 < |z - z_0| < \varepsilon\}$

☐ $\{z \in \mathbb{C} : |z - z_0| > \varepsilon\}$

Evaluate $\lim_{z \rightarrow \infty} \left(\frac{5z^2 + 1}{3z^2 - z} \right)$

Answer



Question 29 (Marks: 3)

Let $S = \{z : |z - 3| \leq 5\}$ be a region. Find its interior points, exterior points and boundary points.

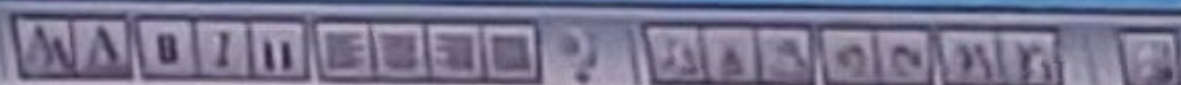
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Question 30 (Marks: 5)

Find $g'(w)$, where $g(w) = \frac{(3+i)w^2}{2w-1}$.

Answer



Question 31 (Marks: 5)

$$f(z) = \frac{z^3}{(z+i)^3} \text{ at } z=i$$

Find the derivative of

Answer

